RAHUL SINGH

Full-Stack Developer · AI/LLM & RAG · Vue.js · Node.js · FastAPI · ERP & LMS

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PROFESSIONAL SUMMARY

AI-focused full-stack developer with 1+ year of production experience building multi-tenant **ERP, LMS, and CRM** platforms with **Vue/Nuxt, Express.js, and MariaDB**, plus a self-built configurable **RAG platform** (FastAPI, LangChain, pgvector). Strong in **RBAC, payment/banking integrations, WhatsApp API, and real-time systems**, with end-to-end ownership from requirements to deployment.

TECHNICAL SKILLS

AI / LLM	RAG, LangChain, pgvector, Embeddings, Chunking, Reranking, Groq, Gemini
Frontend	Vue.js 2, Nuxt.js 2, Vuetify 2, React.js, Next.js, Tailwind CSS
Backend	Node.js, Express.js, FastAPI, REST APIs, WebSockets, SSE, ARQ + Redis, JWT, OAuth 2.0
Databases	MariaDB, PostgreSQL (pgvector), MongoDB, Redis, IndexedDB
Architecture	Multi-Tenant SaaS, RBAC (Entity-Specific Permissions), Org Hierarchy, Offline-First / PWA
Integrations	ICICI Bank API, Razorpay, Subscription Billing, WhatsApp Business API
Languages & Tools	JavaScript ES6+, TypeScript, Python, GitHub, Postman, VS Code

EXPERIENCE

Full Stack Developer (L1) • **Site Guru** *(Promoted from Intern)* Apr 2025 – Present

- Promoted from intern to full-time on the strength of delivery; now **own features end-to-end** across multiple production **multi-tenant SaaS products** (Indicava ERP + LMS, CRM) spanning **Vue.js/Nuxt, Express.js, and MariaDB**, from requirements and schema design to deployment.
- **Led the transformation of Indicava ERP** through a complete UI/UX overhaul and major feature expansion, adding **modular subscriptions**, **fee and finance management**, and an integrated **LMS module with AI-powered answer evaluation**.
- Engineered **payment and banking workflows**: **ICICI Bank API** integration for real-time fee collection, payment ledger, and automated reconciliation, plus **Razorpay** for course billing and subscriptions.
- Built a **CRM with WhatsApp Business API** integration for automated notifications, lead tracking, and customer communication workflows.
- Designed **RBAC with entity-specific permissions and organizational hierarchy**, giving fine-grained access control across roles, departments, and branches; implemented **WebSocket** real-time events and **IndexedDB offline-first sync** for low-connectivity environments.
- Designed **REST APIs**, normalized MariaDB schemas, optimized queries, and built reusable **Vuetify component libraries**, along with AI-assisted analytics and reporting dashboards.
- As a key engineer in a lean startup team, rapidly resolved **urgent production issues** and delivered **custom client requirements**, while owning **end-to-end QA**, bug triage, and release readiness.

PROJECTS

RAG-LAB — Configurable RAG Platform *Personal Project*
Stack: FastAPI, Next.js, LangChain, PostgreSQL (pgvector), ARQ, SSE

- Architected a full-stack **RAG experimentation platform** enabling dynamic, per-thread pipeline orchestration across **query transformation, retrieval, cross-encoder reranking, and citation-grounded generation**.
- Implemented **hybrid retrieval** combining dense vector search and sparse **BM25 keyword search** with weighted fusion, **MMR deduplication**, and **BAAI/bge cross-encoder reranking**.
- Built advanced **query transformation pipelines** supporting **HyDE** (Hypothetical Document Embeddings), multi-query expansion, sub-query decomposition, and step-back prompting to boost retrieval recall.
- Designed modular **knowledge bases** with selectable chunking strategies (*recursive, semantic, markdown, character*) and engineered **asynchronous document ingestion** via **ARQ Redis workers** with real-time SSE progress updates.

Indicava — School SaaS ERP + LMS *2025 – Present*
Stack: Nuxt 2, Vue.js 2, Vuetify 2, Express.js, MariaDB, WebSockets, ICICI Bank API, Razorpay

- Multi-tenant school management platform with admissions, academics, attendance, timetabling, **fee and finance management**, **modular subscriptions**, and **RBAC with entity-specific permissions**.
- Banking module with real-time fee collection, payment ledger, and automated reconciliation via **ICICI Bank API**; **Razorpay** for course enrollment billing.
- Integrated **LMS** (originally built as the standalone BestPrep product): student management, content delivery, enrollments, **live classes**, recording access, and parent/teacher/student/admin roles with granular permissions.
- **Assessment platform**: Test Series and Quiz Series with paper creation, scheduling, auto-evaluation, rank lists, leaderboards, and analytics, plus **AI-powered answer evaluation**.
- **AI-powered insights**: student performance, fee defaulters, attendance trends, and academic progress dashboards.
- Real-time events via **WebSockets**; **offline-first** support via IndexedDB for reliable usage in school environments with patchy connectivity.
- **Super-admin panel**: tenant management, module activation, billing, and gateway config.

EDUCATION

Bachelor of Computer Applications (BCA) Vivekananda Institute of Professional Studies, GGSIPU, New Delhi